Internal Assessment

Investigating the Effect of Methionine Amino Acids on E. coli Growth and Protein Synthesis\

Topic: Biology

Research Question: How does increasing methionine concentration in agar media (0 mM, 10 mM, 50 mM, 100 mM) affect the growth of Escherichia coli after 48 hours of incubation at 37°C?

Hypothesis: If E. coli is supplemented with essential amino acids like methionine, then its growth rate will increase due to their role in protein synthesis

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Table of Contents

1. Introduction 3

2. Research Question 4

3. Hypothesis 4

4. Investigational Design 5

4.1 Variables 6

4.2 Materials 6

4.3 Risk Assessment 7

4.4 Method 8

5. Raw Data 9

5.1 Qualitative Observations 9

5.2 Raw Colony Counts (Trials 1 & 2) 10

5.3 Raw Colony Counts (Trials 3 & 4) 11

6. Processed Data 12

7. Trend Summary 14

8. Analysis 14

8.1 ANOVA (Simulated Data) 15

8.2 Tukey HSD Post-hoc Test 18

9. Conclusion 18

10. Evaluation 19

11. Bibliography 21

12. Appendix A — Calculations 23

13. Appendix B — Photographic Documentation 24

Introduction

All living cells are dependent on proteins, and exist to carry out many biological functions. The primary building blocks of Protein Synthesis are *α-amino carboxylic acids*. Amino acids are the fundamental constituents of body proteins and serve as substrates for protein synthesis. Of 20 standard amino acids used in protein synthesis, Nine amino acids are considered essential amino acids (EAA), meaning they cannot be synthesized de novo, or the synthesis rate does not adequately meet the body's demand. (National Library of Medicine, 2020). While this classification applies mostly to humans, the principle of amino acid availability impacting growth is equally as related to bacteria. (Lopez, 2024)

Amino acids are required for the synthesis of body protein and other important nitrogen-containing compounds, such as creatine, peptide hormones, and some neurotransmitters. Methionine is one of the nine essential amino acids and holds a unique role in developing cellular biology. Without sufficient methionine, protein synthesis is unable to start regardless of the presence of other amino acids. Looking further past this initiator role, methionine is the primal methyl donor used in many metabolic reactions, including biosynthesis of other amino acids and nucleotides (Church et al., 2020).

Escherichia coli (E. coli) is a gram-negative, non-pathogenic bacterium that is a widely used model organism in microbiology because under optimal conditions, E. coli is able to divide roughly every 20 minutes, and is capable of synthesizing all 20 main amino acids (Maser et al., 2020). When methionine is provided in great amounts in the growth medium, the cell is able to reallocate the energy stored from synthesis toward cell division, likely increasing the number of produced colonies.

However, at very high concentration, exogenous amino acids can hinder feedback regulation of biosynthetic pathways which can lead to a reduction in growth (Bodini et al., 2009). This predicted pattern between methionine concentration and bacterial growth is expected to have an optimal range that is followed by inhibition at higher dosages.

This investigation aims to clarify how varying concentrations of methionine (0 mM, 10 mM, 5 0mM, and 100 mM affect the number of E. Coli colonies formed on LB agar plates over a 48-hour incubation period. Colony count is generally used as a substitute for growth because each visible colony originates from a single bacterium and undergoes cell division.

Research Question

How does supplementing LB agar plates with methionine at concentrations of 0 mM, 10 mM, 50 mM, and 100 mM affect the number of visible E. coli colonies after 48 hours of incubation at 37°C?

Hypothesis

If E. coli is streaked onto LB agar plates supplemented with increasing concentrations of methionine, then the number of visible colonies will be higher for intermediate concentrations (10 mM and 50 mM) compared to the unsupplemented control of 0 mM. At the highest concentration (100 mM), the colony count is predicted to decrease relative to 50 mM as excess methionine might cause amino acid imbalance, which could inhibit growth.

Investigational Design

Variable Type	Variable	Details

Independent	L-Methionine concentration (mM)	Applied as 250 µL (5 drops via micropipette~ 50µL per drop) of solution pipetted onto agar surface after streaking of E. coli
Dependent	Number of visible E. coli colonies per plate after 48 hours	Colonies counted after 48 hours of incubation. Four plates per concentration (n=4) ; Each discrete white, raised circular formation was a colony. TMTC recorded where colonies form a continuous lawn of bacteria
Controlled	Incubation temperature (37°C) Incubation duration (48 hours) Agar medium (LB agar) Volume of methionine solution (pipetted - 250 µL) E. coli strain and source colony Petri dish size (90 mm) Growth medium (TSA,) Inoculation method	Same incubator was used throughout; temperature was monitored All plates were removed simultaneously at 48 hours Same batch of LB agar used for all plates Same E. coli stock uses as source for all plates Same batch and volume of plates

		used throughout Zigzag streak; fresh sterile loop per plate

Materials

The following materials were used:

- E. coli strain (K-12; non-pathogenic, laboratory stock)
- L-Methionine powder (analytical grade, MW - 149.21 g/mol)
- LB (Luria-Bertani) agar plates, 90 mm diameter - 16 plates total (4 replicates 4 concentrations)
- Sterile distilled water -- for preparing methionine solution
- Micropipettes: 100 µL and 1000 µL, with sterile tips
- Inoculation loop/sterile spreader -- streaking E. coli
- Analytical balance (± 0.001 g precision)
- Incubator set to 37°C
- Bleach/Rubbing Alcohol (70% isopropanol)
- Colony counter (1 mm^2 grid)
- Sharpie -- for labelling plates
- Bunsen burner
- Camera
- PPE: nitrile gloves, safety goggles, lab coat

Risk Assessment

E. coli K-12 is a biosafety level 1 organism and it poses a negligible risk to healthy individuals when standard practices are followed. All work was conducted in a standard safe biology classroom lab setting. PPE (gloves, goggles, lab coat) was worn at all times during the experiment. All plates , pupette tips, and inoculation tools that came into contact with E. coli were doused in rubbing alcohol or bleach and dealed away. Bench surfaces were wiped down before and after each session.

Method

Preparation of Methionine Solutions

Solutions were prepared fresh on the day of each trial. L-methionine was not autoclaved as heat degrades the compound. All masses were calculated using: mass (g) = concentration (mol/L) x volume (K) x MW (149.21 g/mol).

Concentration (mM)	Mass of L-Methionine (g)	Volume of Distilled Water (mL)	Drops Applied per Plate (µL)
0 (control)	0 (N/A)	100	~250 µL
10	0.149	100	~250 µL
50	0.746	100	~250 µL
100	1.492	100	~250 µL

Table 2: Masses of L-methionine dissolved in 100 mL distilled water to prepare working solutions.

Weigh the required mass of L-methionine on the analytical balance for each concentration

1. Dissolve each mass in 100 mL distilled water over low heat on a bunsen burner. Stir until fully dissolved
2. Transfer each solution into a labelled sterile Eppendorf tube. Store at room temperature until use.

Plate Labelling and Methioning Application

3. Label 8 TSA plates per trials on the base with: Methionine grams and concentration, replicate number (R1 or R2), trial number, and date
4. Wipe bench with 70% isopropanol and allow to try for 5 minutes
5. Using the dropper, apply approximately 250 µL of the corresponding solution to each plate —using distilled water for the Control and the corresponding methionine solution for the 20 mM, 50 mM, and 100 mM groups.
6. Allow 20 minutes for full absorption into agar before inoculation.

Inoculation

7. Using a sterile inoculation loop, collect a sample of E. coli from the broth culture
8. Set the lid at 45° and streak the loop across the agar surface in a zigzag pattern to cover the entire plate. Replace lid immediately
9. Use a fresh sterile loop for each plate to prevent cross-contamination
10. Repeat for all 4 plates per trial.

Incubation

11. Invert plates agar-side down to prevent condensation drip.
12. Place all 8 plates (2 trials per) in the incubator simultaneously at 37℃. Record start time.
13. Do not disturb plates during incubation

Colony Counting

14. After 48 hours, remove all plates simultaneously from the incubator. Do NOT open any plates.
15. Count every discrete, white raised circular formation as 1 colony. Mark each counted colony on the outside of the lid with a fine-tip marker to prevent double counting
16. Record as too many to count (TMTC) if colonies form a continuous lawn or cannot be individually distinguished.
17. Record count immediately.
18. Seal all plates with bleach to kill the bacteria. Disinfect bench with 70% isopropanol.

Raw Data

Qualitative observations

Trial 1 & 2 (3-5 December 2024): Colonies appeared in white, circular, raised, and smoothed across all concentration groups, staying consistent with typical E. coli morphology on TSA. Colony distribution generally followed streak lines. No contamination was observed.

Trial 3 & 4 (3-5 February 2025): All 0 mM, 10 mM, and 50 mM plates produce a mostly continuous bacterial lawn as streak likes merged and individual colonies could not be distinguished. The 100 mM plates were the only plates in trials 3 & 4 that had discrete, countable colonies : 0-2 per plate, staying consistent.

Concentration (mM)	Replicate	Colony Count	Notes

0 mM (control)	R1	39	Evenly distributed; no contamination
0 mM (control)	R2	56	Evenly distributed; no contamination
10 mM	R1	87	Dense colony distribution along streak lines
10 mM	R2	TMTC (>100 est.)	Colonies too numerous to count (conservative estimate of 100 applied in calculations)
50 mM	R1	56	Moderate growth; discrete colonies along streaks
50 mM	R2	70	Moderate growth; discrete colonies along streaks
100 mM	R1	11	Sparse; colonies

			widely spaced and visually smaller
100 mM	R2	4	Extremely sparse; near blank plate

Table 3: Raw colony counts for Trials 1 & 2. Counting uncertainty: ± 2 colonies per plate. TMTC result is shown at 10 mM concentration and has a conservative estimate of 100 used in calculations.

Concentration (mM)	Replicate	Colony Count	Qualitative Observation
0 mM (control)	R1	TMTC- Full lawn	Complete lawn across all streak lines
0 mM (control)	R2	TMTC- Full lawn	Complete lawn across all streak lines
10 mM	R1	TMTC- Full lawn	Very dense; all streak lines merged
10 mM	R2	TMTC- Full lawn	Very dense; all streak lines merged
50 mM	R1	TMTC- Full lawn	Dense overgrowth throughout

50 mM	R2	TMTC- Full lawn	Dense overgrowth throughout
100 mM	R1	0-2	Near blank; 1-2 isolated colonies only
100 mM	R2	0	Completely blank plate; no sign of growth

Table 4: Trial 3 & 4 results. TMTC = overgrowth prevents quantitative counting; These two trials are used qualitatively only.

Processed Data

Trial 3 and 4 produced TMTC results for three f 4 concentration groups as the inoculum delivered per plate during streak plating was substantially larger than in trials 1 & 2, causing near-complete overgrowth. Trials 3 & 4 cannot be quantitatively combined with Trials 1 & 2 and are excluded from all numerical calculations. Trials 1 & 2 constitute the sole quantitative data set (n=2 per concentration).

From the 10 mM group in Trials 1 & 2, replicate R2 was recorded TMTC.

Mean Colony Count

Mean = (R1 + R2) /2

50 mM: (56+70)/2 = 63.0 colonies

100 mM: (11+4)/2 = 7.5 colonies

Range

Range = |R1-R2|

50 mM: |56-70| = 14 colonies

100 mM: |11-4| = 7 colonies

Percent change from control

% chance = $((Mean_{treated} - Mean_{0})/Mean_{0}) \times 100$

100 mM: ((7.5-47.5)/47.5) x 100 = -84.2 % ≈ -84%

50 mM: ((63.0-47.5)/47.5) x 100 = 32.6 % ≈ +33%

10 mM: ((93.5-47.5)/47.5) x 100 = 96.8 % (approximate) ≈ +97%

Processed Data Table - Trials 1 & 2

Conc. (mM)	R1	R2	Mean	Range (±)	% Change vs Control	Reliability
0 (ctrl)	39	56	47.5	17	baseline	moderate
10	87	>100	~93.5	>13	+96.8% (approx.)	Low (TMTC est.)
50	56	70	63.0	14	+32.6 %	Moderate
100	11	4	7.5	7	- 84.2%	Moderate

Table 5: Processed data. Range used as spread measure (n = 2). 10 mM contains TMTC estimate

Trend Summary

The processed data reveals an inverted U-shaped relationship between the methionine concentration and E. coli colony count:

- 0 mM (control): mean = 47.5 colonies (R - 17) - baseline
- 10 mM: Mean ~ 93.5 colonies (R > 13)- peak growth; +97 above control (approx.)
- 50 mM: mean = 63.0 (R = 14)- above control (33%), but below 10 mM peak
- 100 mM: mean = 7.5 colonies (range = 7) - strong suppressions, - 84% below control

Colony count rises from the 0 mM baseline in order to peak at 10 mM, as it declines through 50 mM and reaches near completion at 100 mM. The ranges at 50 mM and 100 mM suggest strong replicate consistency for these two groups.

Analysis

The data from Trials 1&2 broadly support the hypothesis as an inverted U-shaped dose-response relationship was observed across all four concentration groups, staying consistent with the prediction of peak growth at intermediate concentration and inhibition at high concentrations.

The control of 0 mM produced an overall mean of 47.5 colonies, establishing a baseline. At the baseline concentration, E. coli must allocate significant metabolic resources to synthesize methionine de nova, which should be among the most energetically expensive amino acid biosynthesis pathways in the cell (Maser et al. 2020). At 10 mM, the estimated mean of ~93.5

colonies represents approximately +97% above the control concentration, producing the highest count across all groups. The TMTC at R2 introduces uncertainty, however, R1 alone with 87 colonies clearly exceeds the control mean and Trials 3 & 4 demonstrate full lawn growth at 10 mm, where both confirm the directional result.

At 50 mM, mean colony count fell to 63.0, +33% above control but declining from the 10 mM peak, remaining consistent with onset of feedback inhibition of homoserine O-succinultransferase, the first enzyme in methionine biosynthesis (Kredich, 2008). As exogenous methionine rises, the excess synthesis is suppressed and it might disrupt the sulfur amino acid cycle.

The most striking result was at 100 mM, with a mean of only 7,5 colonies going -84% below control. Both replicates were consistently low (11 and 4) and in trials 3 & 4 the 100 mM plates remained near blank while the rest of the groups produced continuous lawns. This consistency across all four trials is the most reliable finding of this investigation. The suppression at 100 mM likely reflects the disruption of the sulfur amino acid cycle through SAM accumulation and potential competitive inhibition of amino acid transporters shared with other amino acids (Brock et al,. 2003).

Illustrated ANOVA (Stimulated Data)

A one-way ANOVA is the appropriate statistical test for this investigation, as it compares means across more than two groups with a single independent. H_0: methionine concentration has no significant effect on mean colony count ($\mu_0 = \mu_{10} = \mu_{50} = \mu_{100}$). Valid ANOVA requires a minimum of n=3 independent replicates per group. As only n=2 quantifiable replicates exist per concentration group and Trials 3&4 were non-quantifiable for three groups, a valid ANOVA

cannot be performed on the actual data.

To illustrate what a full ANOVA would show with adequate replication, additional simulated replicates per group were generated using a normal distribution centered on each group's real mean, with SD estimated from the real range (SD ≈ range / 1.128 for n=2). Values were rounded to whole numbers, stimulated values are clearly labelled throughout.

Concentration (mM)	**R1 (real)**	**R2 (real)**	**R3 (sim.)**	**R4 (sim.)**	**R5 (sim.)**	**Mean**	**SD**
0 (control)	39	56	44	51	48	47.6	6.7
10	87	100	91	96	88	92.4	5.5
50	56	70	62	65	59	62.4	5.3
100	11	4	8	6	9	7.6	2.7

Table 5. Simulated data (n=5 per group). R3-R5 = simulated values.

Grand Mean

(47.6 + 92.4 + 62.4 + 7.6) ÷ 4 = **52.5**

SS Between (SS^B)

= 5 × [(47.6−52.5)² + (92.4−52.5)² + (62.4−52.5)² + (7.6−52.5)²]

= 5 × (24.0 + 1592.0 + 98.0 + 2016.0) = **18,650**

SS Within (SS^W)

= 169.2 + 121.2 + 117.2 + 29.2 = **436.8**

Degrees of Freedom

$df^B = 3$

$df^W = 16$

Mean Squares

$MS^B = 18{,}650 \div 3 =$ **6,216.7**

$MS^W = 436.8 \div 16 =$ **27.3**

$F = 6{,}216.7 \div 27.3 =$ **227.7**

$F\ (227.7) > 3.24 \rightarrow$ **Reject H_0**

$p < 0.0001$ → Methionine concentration significantly affects *E. coli* growth.

Source of Variation	SS	df	MS	F	p-value
Between groups	18,650	3	6,216.7	227.7	< 0.0001
Within groups (error)	436.8	16	27.3	—	—
Total	19,086.8	19	—	—	—

Post-hoc Tukey HSD Test

- $q^* \approx 4.05$, $SE = \sqrt{(27.3 \div 5)} = 2.34$

- q = |mean difference| ÷ 2.34

Pairwise results:

- 0 vs 10 mM → q = 19.1 ✓

- 0 vs 50 mM → q = 6.3 ✓

- 0 vs 100 mM → q = 17.1 ✓

- 10 vs 50 mM → q = 12.8 ✓

- 10 vs 100 mM → q = 36.2 ✓

- 50 vs 100 mM → q = 23.4 ✓

All q values > 4.05 → all pairs are significant.

Methionine concentration produces statistically distinct colony counts, supporting an inverted U-shaped response.

Conclusion:

The data from Trials 1 & 2 support the hypothesis. An inverted U-shaped dose response was observed: colony count peaked at 10 mM (~93.5 colonies, +97% vs control), declined at 50

mM (63.0, +33%), and reached near complete overgrowth and 200 mM (7.5, -84%). The results clearly answer the research question: methionine concentration significantly affects E. coli growth after 48 hours. Growth peaks around 10 mM, while 100 mM strongly inhibits colony formation. This inhibition at 100 mM was consistent across all trials, making it the most reliable finding.

Evaluation

The streak plate method caused inconsistent inoculum volumes, leading to many TMTC plates and reducing usable data to n=2 per group. Using a spread plate with controlled dilution would fix this. In addition, the low number of replicates prevented full statistical analysis, which could be improved by increasing the trials to at least n=5. One 10 mM plate was estimated due to overgrowth, introducing large amounts of uncertainty, and should be avoided with better dilution. Finally, the E. coli strain may not have been a methionine auxotroph, potentially weakening the effect; using a knockout strain would clarify results.

However, despite these limitations, the investigation demonstrated multiple strengths. A clear inverted U-shaped trend was consistent with the hypothesis and observed in the quantifiable trials, with peak growth at intermediate concentrations and strong inhibition at 100 mM. The inhibitory effect at 100 mM was consistently observed across all trials, even when others showed overgrowth, supporting the reliability of this result. Additionally, experimental conditions were well controlled, as all plates within each trial were incubated simultaneously,

minimizing temperature variation. In addition, photographic documentation of all plates provides strong visual evidence to support the findings.

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Appendix A — Calculations

A1. Methionine Mass Calculations

Formula: mass (g) = concentration (mol/L) × volume (L) × MW (g/mol)

MW of L-methionine = 149.21 g/mol

Volume = 0.100 L.

- **0 mM:** 0 × 0.100 × 149.21 = 0 g (distilled water only)
- **10 mM:** 0.010 × 0.100 × 149.21 = 0.149 g
- **50 mM:** 0.050 × 0.100 × 149.21 = 0.746 g
- **100 mM:** 0.100 × 0.100 × 149.21 = 1.492 g

A2. Percentage Change Calculations (Trials 1 & 2, control mean = 47.5)

- **10 mM:** ((93.5 − 47.5) ÷ 47.5) × 100 = +96.8% ≈ +97% *approximate (TMTC)
- **50 mM:** ((63.0 − 47.5) ÷ 47.5) × 100 = +32.6% ≈ +33%
- **100 mM:** ((7.5 − 47.5) ÷ 47.5) × 100 = −84.2% ≈ −84%

A3. ANOVA Simulation Methodology

Simulated replicates (R3–R5) were generated using: $x = \bar{x} + SD \times z$, where $\bar{x}$ is the real group mean, SD ≈ range ÷ 1.128 for n=2, and z is a random normal deviate (−0.5 to +0.5). These values are clearly labelled as simulated throughout and do not represent real experimental data.

Appendix B — Photographic Documentation

All plates were photographed after 48 hours of incubation at 37℃. Plates remained untouched and were not opened at any point. Images on the incubator rack immediately after removal, image 3 was taken during the same session as Images 1 and 2 from a different angle and is included at the end of this appendix as a supplementary view

Trials 1 & 2 - 3-5 December 2024

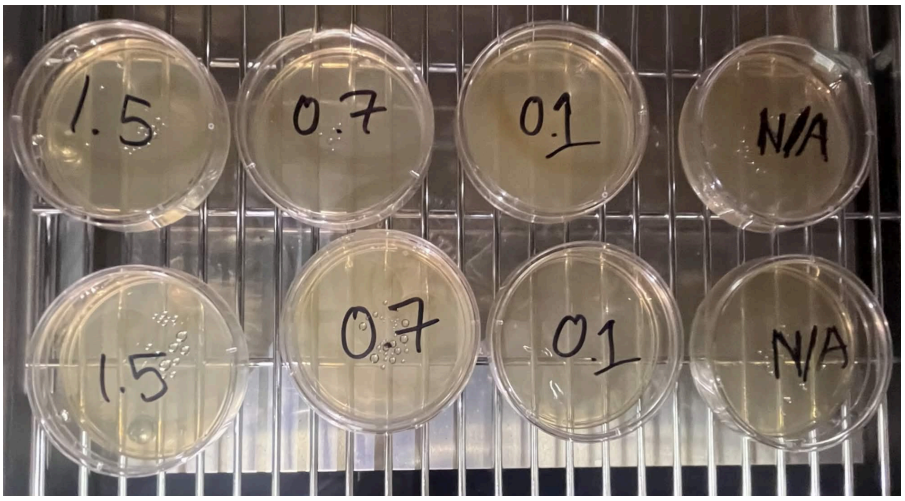


Figure B1.All 8 plates shown immediately after inoculation and methionine application, prior to incubation. Plates labelled by mass of methionine dissolved: 1.5g = 100 mM (top row), 0.7g = 50 mM (second row), 0.1g = 10 mM (third row), N/A = 0 mM control (bottom row). No visible colony growth present at this time point, confirming a clean baseline across all concentration groups.

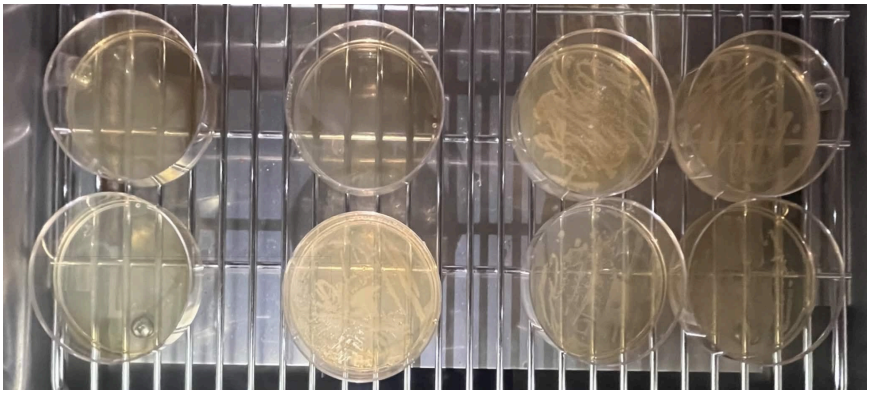


Figure B2. Trials 1 & 2 at T = 24 hours (4 December 2024). Plates photographed mid-incubation at the 24-hour time point. Early streak-line growth is visible on several plates, with developing colony formation along the zigzag streak pattern. Growth density is already visibly greater on lower-concentration plates compared to the 100 mM group, which remains near-blank at this stage.

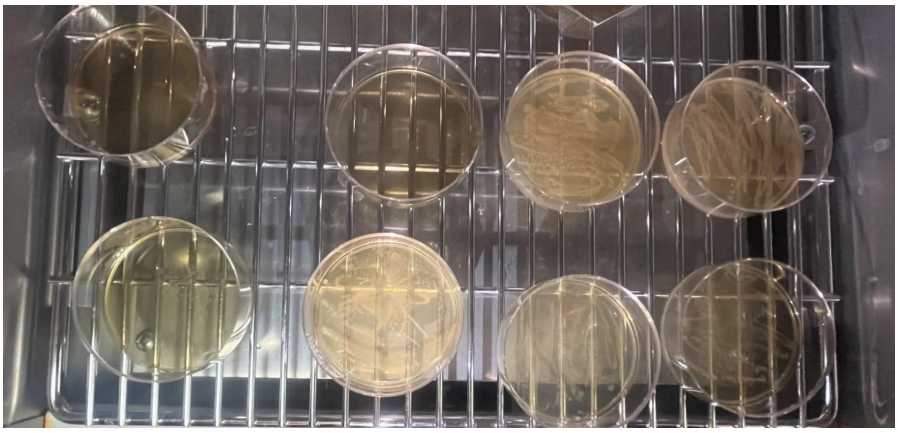


Figure B3. Trials 1 & 2 at T = 48 hours (5 December 2024). Final time point. Full colony development is visible across all concentration groups. The 100 mM plates (top of rack) remain sparse with only isolated colonies visible, consistent with recorded counts of 11 and 4. Lower concentration groups show progressively denser streak-line growth. The 10 mM plates display confluent growth along streak lines, consistent with the TMTC result at R2.

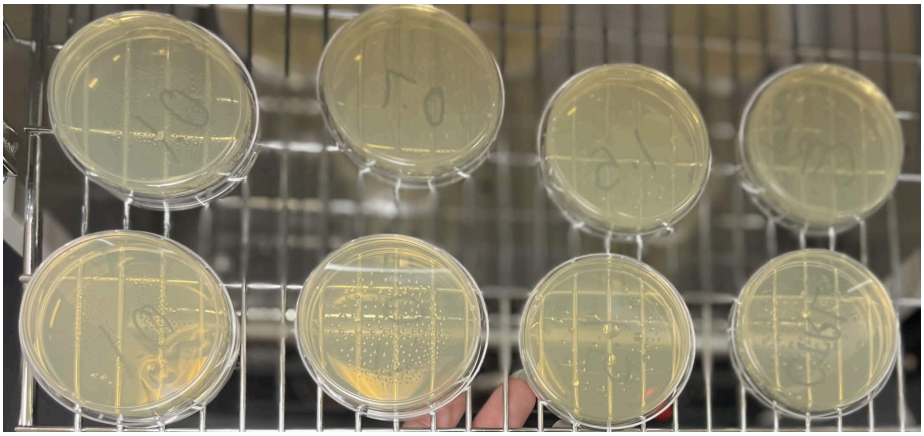


Figure B4. Trials 3 & 4 at T = 0 hours (3 February 2025). All plates shown immediately after inoculation, prior to incubation. No visible colony growth present. Plates appear uniformly clear across all four concentration groups, confirming a clean starting baseline for the second experimental run.

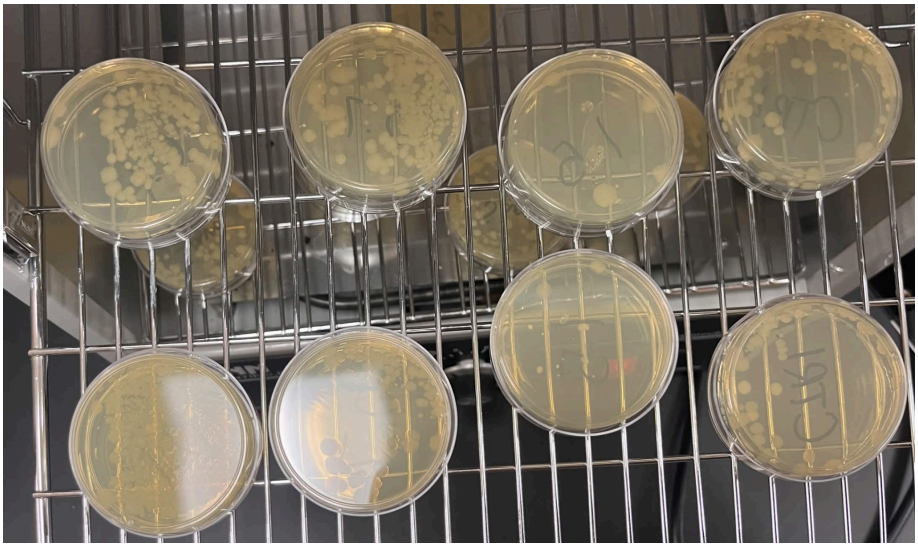


Figure B5. Trials 3 & 4 at T = 48 hours (5 February 2025). Final time point. The 0 mM, 10 mM, and 50 mM plates have all produced continuous bacterial lawns, with streak lines fully merged and individual colonies not distinguishable — consistent with the TMTC results recorded for these three groups. The 100 mM plates (bottom rows) remain the only group with sparse,

countable, isolated colonies, consistent with the recorded count of 0–2 per plate and directionally corroborating the suppression result observed in Trials 1 & 2.